

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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CIVIL / STRUCTURAL CALCULATIONS

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Revision History

Rev No.	Date	Description of Revision
R01	29-06-22	Issued for Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-CIV-RPT-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

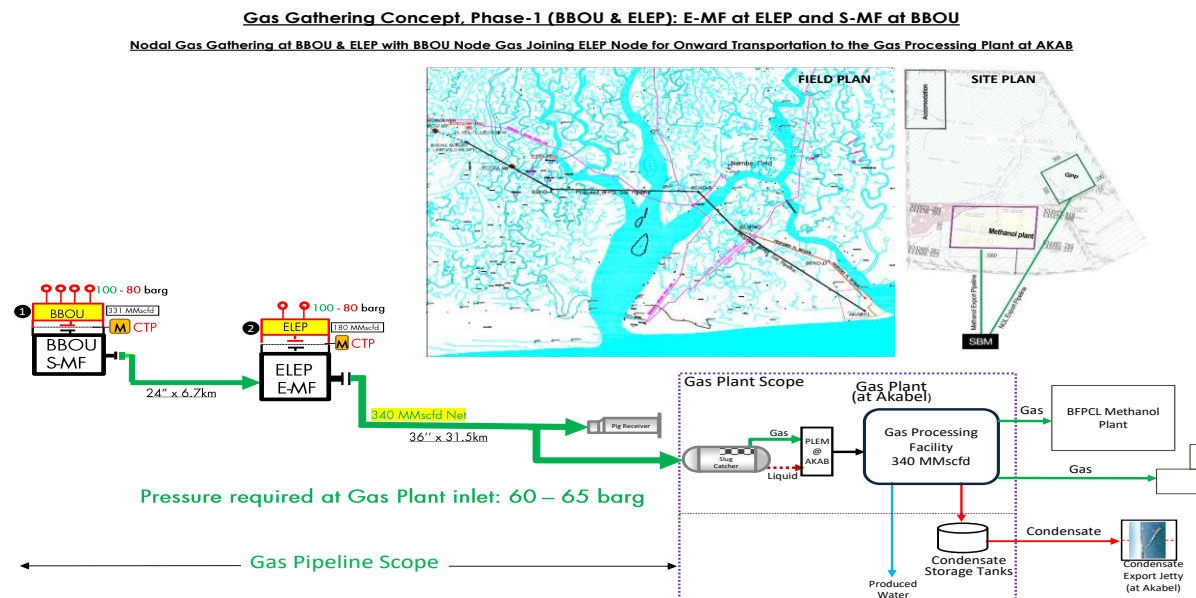


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purposes of this analysis report is to evaluate the structural adequacy of the manifold platforms and to ascertain their ability to sustain the loads that may occur during in-service conditions. This report contains the following analyses:

Elepa Manifold Platform

Boubouwei Manifold Platform

Structural model of the third manifold Akabel, is not included as it is going to be supported on reinforced concrete plinth.

1.3. Conclusion

- The Manifold Platforms meet the requirements described in the Design Basis during the In-place conditions.
- The stress ratios of structural beams and columns do not exceed 1.0 which is the allowable. The maximum utilisation ratio obtained is 0.93
- The displacements are found acceptable.

1.4. Models Descriptions

The Elepa and Boubouwei Manifold platforms have two levels each at elevations +3000mm and + 6000mm respectively above ground level. They are supported by steel columns connected to pile caps. The lower level houses the equipment and the piping whereas the upper level is used to accommodate monorail for lifting and loading the barrels of the pig receiver and launcher.

In this analysis, considering the massive nature of the platform foot print, only the primary steel members and the secondary steel are modelled. The tertiary members and grating were calculated and distributed on the primary members and are referred to as un-modeled steel members.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Design Criteria

3.1. Steel Grade

The tensile strength and yield strength are listed below in MPa

Table 3a. Tensile & Yield strength

Grade	Minimum Tensile strength f_u (MPa)	Minimum Yield strength F_y (MPa)		
		$t < 16\text{mm}$	$16 < t \leq 40\text{mm}$	$40 < t \leq 63\text{mm}$
	470	248	345	335

Mechanical properties:

- Young's modulus: $E = 210000 \text{ MPa}$
- Poisson's ratio: $= 0.3$
- Density: $= 7.850 \text{ mT/m}^3$
- Thermal expansion: $= 12.0 \times 10^{-6} / ^\circ\text{C}$

3.2. Deflections

According to section 6.4 of the Standard structural specification (ref. [3]), the vertical deflection of the beams shall be checked to ensure the following limits:

Table 3b. Deflection criteria

Condition	$d_{\max}$
Floor beam	$L/240$
Floor Beam support Architectural	$L/360$
Cantilever Floor beam	$L/120$

* Where L = span of the beam

4. Computer Models

The 3D general view of the Inplace Analysis model for Elepa and Bubouwe platforms are shown in the figures below.

4.1. General View

ELEPA PLATFORM

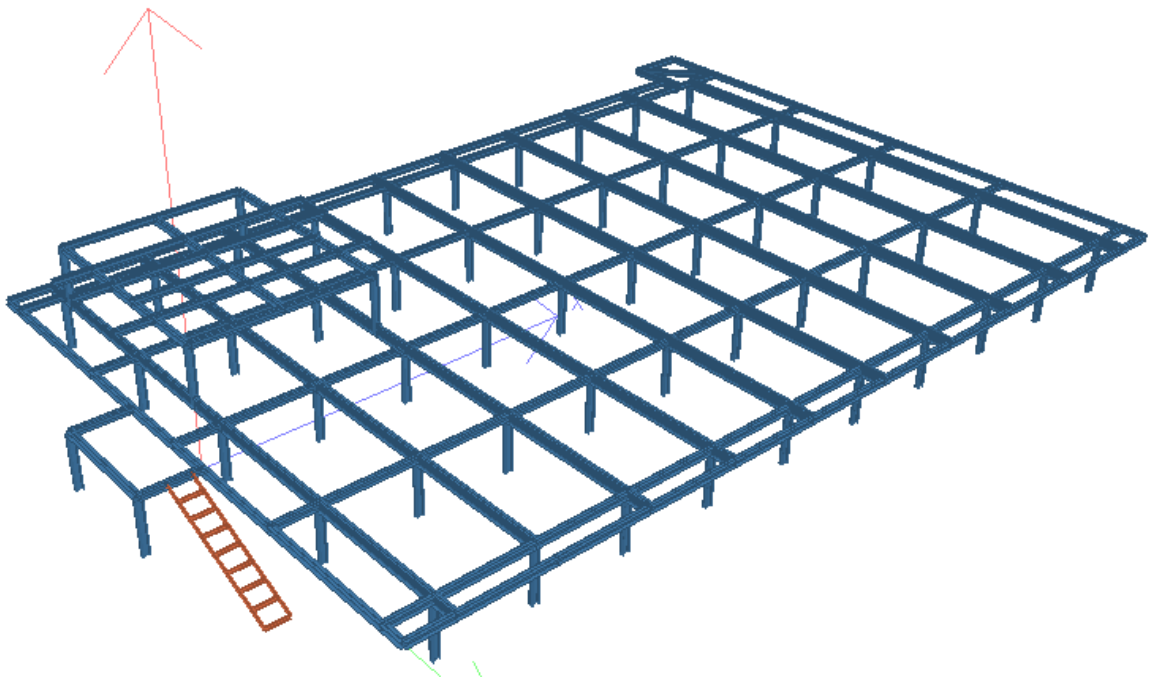


Fig 1: A three-dimensional STAAD. PRO Model of ELEPA Manifold Platform.

BUBOUWE PLATFORM

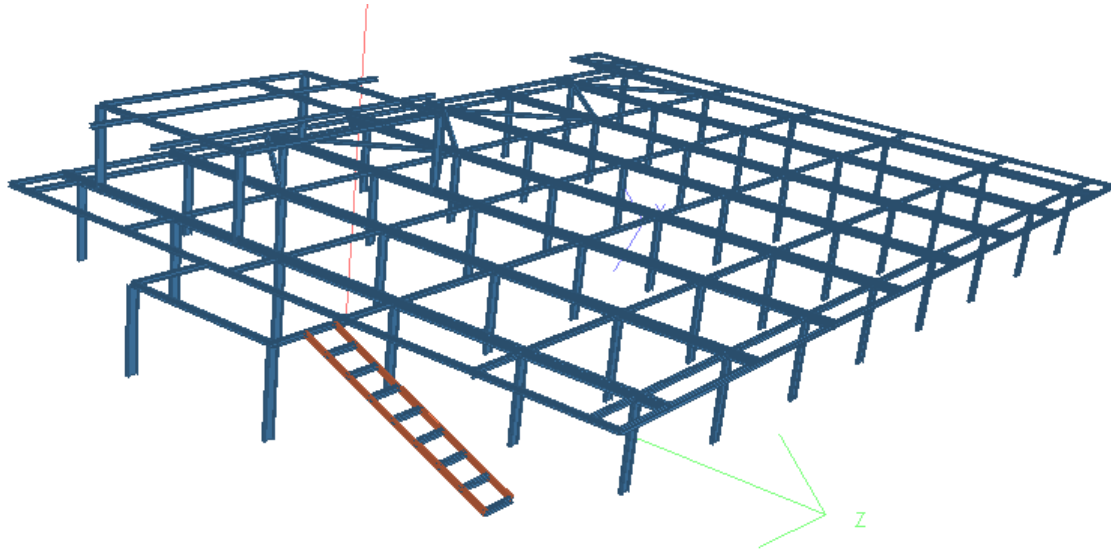


Fig 2: A three-dimensional STAAD. PRO Model of Bubouwe Manifold Platform.

4.2. Member Properties

All member properties as used in analysis are detailed in Table 4a below.

Table 4a: Member Profile

Platform Location	Section Name	Width mm	Flange thk mm	Height mm	Web thk mm
ELEPA	W12X79	307	19	314	12
ELEPA	W14X48	204	15	350	9
ELEPA	W24X94	230	22	617	13
ELEPA	C8X13	60	10	203	8
BUBOUWE	W6X20	153	9	157	7
BUBOUWE	W6X25	154	12	162	8
BUBOUWE	W12X45	204	15	306	9
BUBOUWE	W14X22	127	9	349	6
BUBOUWE	C8X13	60	10	203	8

5. Loadings

5.1. Weight Report

The weights presented for these analysis are mainly that of the piping loads, other loads such as mechanical, electrical, Shell manifold loads are assumed to be twenty percentage of the piping loads.

For Boubouwei platform the pipe load is as presented below

NPS	DESCRIPTION	QTY	WEIGHT (kg)	TOTAL WEIGHT (kg)	
24"	24 " PIPE	110.466	141	15575.706	
8"	8" PIPE	7.071	43	304.053	
6"	6" PIPE	10.985	28	307.58	
4"	4" PIPE	21	16	336	
2"	2" PIPE	24.285	5	121.425	
24"	#900 BALL VALVE	12	6342.89	76114.68	
8"	#900 BALL VALVE	4	609.09	2436.36	
6"	#900 BALL VALVE	4	331.68	1326.72	
6"	#150 BALL VALVE	4	109.29	437.16	
4"	#900 BALL VALVE	11	146.46	1611.06	
4"	#900 GLOBE VALVE	4	146.71	586.84	
2"	#900 BALL VALVE	32	61.49	1967.68	
2"	#900 GLOBE VALVE	10	64.67	646.7	
3/4"	#900 BALL VALVE	10	2	20	
3/4"	#150 BALL VALVE	6	2	12	
24"	#900 RTJ FLANGE	34	1131.26	38462.84	
8"	#900 RTJ FLANGE	6	103.73	622.38	
6"	#900 RTJ FLANGE	4	72.72	290.88	
6"	#150 RTJ FLANGE	13	14.81	192.53	
4"	#900 RTJ FLANGE	27	37.01	999.27	
2"	#900 RTJ FLANGE	31	18.9	585.9	
3/4"	#900 SW FLANGE	10	4.3	43	
3/4"	#900 SW FLANGE	6	1.49	8.94	
24" X 24"	EQUAL TEE	6	603.41	3620.46	
24" X 8"	WELD OLET	2	16.78	33.56	
24" X 6"	WELD OLET	1	10.3	10.3	
24" X 4"	WELD OLET	1	2.9	2.9	
24" X 2"	WELD OLET	11	0.89	9.79	
8" X 4"	RED. TEE	2	21.73	43.46	
8" X 3/4"	WELD OLET	2	0.14	0.28	
6" X 6"	EQUAL TEE	6	19.69	118.14	

6" X 4"	RED. TEE	3	12.16	36.48	
6" X 2"	WELD OLET	2	0.79	1.58	
4" X 4"	EQUAL TEE	5	7.97	39.85	
4" X 2"	RED. TEE	2	4.91	9.82	
4" X 3/4"	WELD OLET	4	0.14	0.56	
6" X 3/4"	WELD OLET	4	0.14	0.56	
24"	#900 RTJ BLIND FLANGE	1	1326.26	1326.26	
2"	#900 RTJ BLIND FLANGE	10	4.1	41	
3/4"	#900 RTJ BLIND FLANGE	10	4.75	47.5	
3/4"	#150 RTJ BLIND FLANGE	6	4.75	28.5	
24"	90 ° ELBOW	10	624.76	6247.6	
8"	90 ° ELBOW	10	30.44	304.4	
6"	90 ° ELBOW	12	14.65	175.8	
4"	90 ° ELBOW	14	5.29	74.06	
2"	90 ° ELBOW	21	0.88	18.48	
24"	INSULATION JOINT	2	720	1440	
				156622.564	157T

The piping loads for Elepa manifold is shown in the table below.

S/NO	NPS	DESCRIPTION		QTY	WEIGHT (kg)	TOTAL WEIGHT (kg)	
1	36"	36 " PIPE		90.35	212	19154.2	
2	24"	24 " PIPE		42.218	141	5952.738	
3	12"	12 " PIPE		27.056	74	2002.144	
4	8"	8" PIPE		14.566	43	626.338	
5	6"	6" PIPE		83.653	28	2342.284	
6	4"	4" PIPE		107.35 1	16	1717.616	
7	2"	2" PIPE		33.075	5	165.375	
8	36"	#900 BALL VALVE		11	26809	294899	
9	24"	#900 BALL VALVE		3	6342.89	19028.67	
10	12"	#900 BALL VALVE		4	1341.87	5367.48	
11	8"	#900 BALL VALVE		2	609.09	1218.18	
12	6"	#900 BALL VALVE		8	331.68	2653.44	
13	6"	#150 BALL VALVE		7	109.29	765.03	
14	4"	#900 BALL VALVE		16	146.46	2343.36	

15	4"	#900 GLOBE VALVE		4	146.71	586.84	
16	2"	#900 BALL VALVE		34	61.49	2090.66	
17	2"	#900 GLOBE VALVE		6	64.67	388.02	
18	3/4"	#900 BALL VALVE		5	2	10	
19	3/4"	#150 BALL VALVE		3	2	6	
20	36"	#900 RTJ FLANGE		25	2781.36	69534	
21	24"	#900 RTJ FLANGE		5	1131.26	5656.3	
22	12"	#900 RTJ FLANGE		10	217.91	2179.1	
23	8"	#900 RTJ FLANGE		3	103.73	311.19	
24	6"	#900 RTJ FLANGE		14	72.72	1018.08	
25	6"	#150 RTJ FLANGE		16	14.81	236.96	
26	4"	#900 RTJ FLANGE		31	37.01	1147.31	
27	2"	#900 RTJ FLANGE		24	18.9	453.6	
28	3/4"	#900 SW FLANGE		6	4.3	25.8	
29	3/4"	#900 SW FLANGE		6	1.49	8.94	
30	36" X 36"	EQUAL TEE		3	463.27	1389.81	
31	36" X 16"	RED. TEE		2	27.67	55.34	
32	36" X 8"	WELD OLET		2	16.78	33.56	
33	36" X 6"	WELD OLET		2	10.3	20.6	
34	36" X 4"	WELD OLET		1	2.9	2.9	
35	36" X 2"	WELD OLET		14	0.79	11.06	
36	36" X 3/4"	WELD OLET		3	0.14	0.42	
37	24" X 24"	EQUAL TEE		1	603.41	603.41	
38	24" X 8"	WELD OLET		1	16.78	16.78	
39	24" X 6"	WELD OLET		1	10.3	10.3	
40	24" X 4"	WELD OLET		1	2.9	2.9	
41	24" X 2"	WELD OLET		11	0.89	9.79	
42	12" X 4"	WELD OLET		2	2.9	5.8	
43	12" X 2"	WELD OLET		4	0.79	3.16	

44	12" X 3/4"	WELD OLET		2	0.14	0.28	
45	8" X 4"	RED. TEE		1	21.73	21.73	
46	8" X 2"	WELD OLET		2	0.79	1.58	
47	8" X 3/4"	WELD OLET		2	0.14	0.28	
48	6" X 6"	EQUAL TEE		7	19.69	137.83	
49	6" X 4"	RED. TEE		7	12.16	85.12	
50	6" X 2"	WELD OLET		2	0.79	1.58	
51	4" X 4"	EQUAL TEE		7	7.97	55.79	
52	4" X 2"	RED. TEE		2	4.91	9.82	
53	4" X 3/4"	WELD OLET		4	0.14	0.56	
54	6" X 3/4"	WELD OLET		3	0.14	0.42	
55	36"	#900 RTJ BLIND FLANGE		2	2753.36	5506.72	
56	2"	#900 RTJ BLIND FLANGE		10	18.9	189	
57	3/4"	#900 RTJ BLIND FLANGE		10	4.75	47.5	
58	3/4"	#150 RTJ BLIND FLANGE		6	4.75	28.5	
59	36"	90 ° ELBOW		8	640.47	5123.76	
60	24"	90 ° ELBOW		3	624.76	1874.28	
61	12"	90 ° ELBOW		9	92.27	830.43	
62	8"	90 ° ELBOW		5	30.44	152.2	
63	6"	90 ° ELBOW		16	14.65	234.4	
64	4"	90 ° ELBOW		36	5.29	190.44	
65	6"	45 ° ELBOW		2	5	10	
66	2"	90 ° ELBOW		27	0.88	23.76	
67	36"	INSULATION JOINT		2		0	
68	24"	INSULATION JOINT		1	720	720	
						459300.465	460T

5.2. Elementary Load Cases

Design loads used in this analysis include Functional loads such as dead, equipment loads and environmental loads.

Table 5a: Elementary Load cases

S/N	Load Description	Applied Value (Tonnes)		Load Type
		BOUBOUWEI	ELEPA	
1	Piping Dry Weight (Bulk)	157	460	Live
2	Modeled Weight (self-weight)	STAAD Generated	STAAD Generated	Dead
3	Un-modeled (Grating/un-modeled steel) 1% of Piping load	1.57	4.60	Live
4	Equipment Dry Weight (10% of Piping)	15.7	46.0	Live
5	Wind load (Negligible considering elevation and terrain)	0.0001	0.0001	Wind

5.3. Software Used

The computer program used for this structural design is:

STAAD PRO V8i – Developed by BENTLEY.

6. Summary Results

6.1. Elepa Manifold Platform

Below are the reactions at supports

Table 6a: Reaction Forces

Reaction Summary

	Node	L/C	Horizontal FX (Mton)	Vertical FY (Mton)	Horizontal FZ (Mton)	Moment		
						MX (kip'in)	MY (kip'in)	MZ (kip'in)
Max FX	112	6:COMBNATIO	0.859	20.093	0.288	0.000	0.000	0.000
Min FX	122	6:COMBNATIO	-0.526	3.644	0.028	0.000	0.000	0.000
Max FY	134	6:COMBNATIO	-0.033	46.913	-0.016	0.000	0.000	0.000
Min FY	93	2:LIVE LOAD	-0.041	-0.134	0.003	0.000	0.000	0.000
Max FZ	152	6:COMBNATIO	0.001	0.036	0.344	0.000	0.000	0.000
Min FZ	151	6:COMBNATIO	0.001	0.503	-0.757	0.000	0.000	0.000
Max MX	93	1:DEAD	-0.219	5.803	0.026	0.000	0.000	0.000
Min MX	93	1:DEAD	-0.219	5.803	0.026	0.000	0.000	0.000
Max MY	93	1:DEAD	-0.219	5.803	0.026	0.000	0.000	0.000
Min MY	93	1:DEAD	-0.219	5.803	0.026	0.000	0.000	0.000
Max MZ	93	1:DEAD	-0.219	5.803	0.026	0.000	0.000	0.000
Min MZ	93	1:DEAD	-0.219	5.803	0.026	0.000	0.000	0.000

6.1.1. Deflections

Table 6b: Displacement at nodes

Node Displacement Summary

	Node	L/C	X (in)	Y (in)	Z (in)	Resultant (in)	rX (rad)	rY (rad)	rZ (rad)
Max X	170	6:COMBNATIO	0.056	-0.156	0.002	0.166	-0.001	-0.000	0.000
Min X	169	6:COMBNATIO	-0.049	-0.035	-0.019	0.063	-0.000	0.001	-0.000
Max Y	49	6:COMBNATIO	0.009	0.096	0.017	0.098	0.002	-0.000	0.000
Min Y	194	6:COMBNATIO	0.032	-0.851	0.014	0.852	-0.001	-0.000	-0.000
Max Z	195	6:COMBNATIO	0.034	-0.790	0.031	0.792	0.001	-0.000	0.000
Min Z	155	6:COMBNATIO	0.003	-0.240	-0.100	0.260	-0.000	0.000	-0.000
Max rX	171	6:COMBNATIO	0.039	-0.013	0.002	0.042	0.004	-0.000	-0.001
Min rX	173	6:COMBNATIO	0.028	-0.038	0.000	0.047	-0.004	-0.000	-0.001
Max rY	169	6:COMBNATIO	-0.049	-0.035	-0.019	0.063	-0.000	0.001	-0.000
Min rY	170	6:COMBNATIO	0.056	-0.156	0.002	0.166	-0.001	-0.000	0.000
Max rZ	92	6:COMBNATIO	0.001	-0.087	-0.025	0.091	0.000	-0.000	0.001
Min rZ	13	6:COMBNATIO	0.009	-0.168	0.018	0.169	-0.004	-0.000	-0.002
Max Rst	194	6:COMBNATIO	0.032	-0.851	0.014	0.852	-0.001	-0.000	-0.000

The Vertical maximum displacement is as shown above. The value is 0.85inch = 21.60mm which occurs at joint 194

Vertical Deflections at Deck. Elevation +6000.00

Member Length = 5000mm

Displacement = 21.60mm

Allowable Displacement = $L/120 = 5000/120 = 41.67$

Therefore, the deflection is less than the allowable. OK

6.1.2. Utilization Ratio

Table 6c: Members Maximum Utilization Check

Sections	Member	UC ratio
W12X79	211	0.32
W14X48	244	0.92
W24X94	89	0.93
C8X13	227	0.76

The allowable utilization Ratio is unity. All the section utilization ratio is below this value.

6.2. Boubouwe Manifold Platform

Table 6d: Reaction Forces

Reaction Summary

	Node	L/C	Horizontal	Vertical	Horizontal	Moment		
			FX (Mton)	FY (Mton)	FZ (Mton)	MX (kip'in)	MY (kip'in)	MZ (kip'in)
Max FX	252	6:COMBNATIO	0.283	8.709	0.056	0.000	0.000	0.000
Min FX	277	6:COMBNATIO	-0.197	0.747	0.008	0.000	0.000	0.000
Max FY	263	6:COMBNATIO	-0.015	22.669	-0.001	0.000	0.000	0.000
Min FY	279	2:LIVE LOAD	-0.000	-0.829	0.003	0.000	0.000	0.000
Max FZ	152	6:COMBNATIO	0.000	0.181	0.161	0.000	0.000	0.000
Min FZ	151	6:COMBNATIO	-0.001	0.372	-0.285	0.000	0.000	0.000
Max MX	151	1:DEAD	-0.001	0.231	-0.172	0.000	0.000	0.000
Min MX	151	1:DEAD	-0.001	0.231	-0.172	0.000	0.000	0.000
Max MY	151	1:DEAD	-0.001	0.231	-0.172	0.000	0.000	0.000
Min MY	151	1:DEAD	-0.001	0.231	-0.172	0.000	0.000	0.000
Max MZ	151	1:DEAD	-0.001	0.231	-0.172	0.000	0.000	0.000
Min MZ	151	1:DEAD	-0.001	0.231	-0.172	0.000	0.000	0.000

6.2.1. Deflections

Table 6e: Deflections at nodes

Node Displacement Summary

	Node	L/C	X (in)	Y (in)	Z (in)	Resultant (in)	rX (rad)	rY (rad)	rZ (rad)
Max X	307	6:COMBNATIO	0.042	-0.195	0.002	0.200	-0.001	-0.000	0.001
Min X	309	6:COMBNATIO	-0.095	-0.198	0.056	0.226	-0.006	-0.000	0.002
Max Y	170	2:LIVE LOAD	0.001	0.256	0.017	0.256	0.002	-0.000	-0.001
Min Y	317	6:COMBNATIO	-0.087	-1.010	0.040	1.015	-0.004	-0.000	0.002
Max Z	320	6:COMBNATIO	-0.087	-0.422	0.064	0.435	-0.004	-0.000	0.003
Min Z	155	6:COMBNATIO	0.003	-0.366	-0.155	0.398	-0.000	0.000	-0.001
Max rX	310	6:COMBNATIO	-0.071	-0.495	0.040	0.502	0.007	-0.000	0.002
Min rX	308	6:COMBNATIO	-0.093	-0.036	-0.026	0.103	-0.007	-0.000	-0.000
Max rY	148	6:COMBNATIO	0.012	-0.064	-0.028	0.070	0.003	0.001	-0.002
Min rY	213	6:COMBNATIO	0.009	-0.015	0.002	0.018	-0.002	-0.000	-0.000
Max rZ	313	6:COMBNATIO	-0.087	-0.668	0.056	0.676	-0.004	-0.000	0.003
Min rZ	169	6:COMBNATIO	0.020	-0.621	0.041	0.623	-0.002	-0.000	-0.008
Max Rst	317	6:COMBNATIO	-0.087	-1.010	0.040	1.015	-0.004	-0.000	0.002

The Vertical maximum displacement is as shown above. The value is 1.0 inch = 25.40mm which occurs at joint 317

Vertical Deflections at Deck. Elevation +6000.00

Member Length = 7000mm

Displacement = 25.40mm

Allowable Displacement = $L/120 = 7000/120 = 58.33$

Therefore, the deflection is less than the allowable. OK

6.2.2. Utilization Ratio

Table 6f: Members Maximum Utilization Check

UTILIZATION RATIOS

Sections	Member	UC ratio
W6X20	376	0.34
W6X25	447	0.37
W12X45	287	0.78
W14X22	435	0.91
C8X13	217	0.2

The allowable utilization Ratio is unity. All the section utilization ratio is below this value.

6.3. Bubouwe Typical Pile cap (Indicative)

File Cap P151 **PILE ARRANGEMENT**

Column Dimensions

Column Shape : Rectangular
 Column Length - X (Pl) : 0.305 m
 Column Width - Z (Pw) : 0.305 m

Pedestal

Include Pedestal? No
 Pedestal Shape : N/A
 Pedestal Height (Ph) : N/A
 Pedestal Length - X (Pl) : N/A
 Pedestal Width - Z (Pw) : N/A

Pile Cap Geometrical Data

Pile Cap Length P_{CL} = 1.400 m
 Pile Cap Width P_{CW} = 0.400 m
 Initial Pile Cap Thickness t_1 = 0.300 m

Pile Geometrical Data

Pile spacing P_s = 1.000 m
 Pile Edge distance a = 0.200 m
 Pile Diameter d_p = 0.300 m

Pile Capacities

Axial Capacity P_p = 215.746 kN
 Lateral Capacity P_L = 9.807 kN
 Uplift Capacity P_U = 0.000 kN

Material Properties

Concrete f'_c = 25000.004 kN/m²
 Reinforcement f_y = 415000.070 kN/m²

Concrete Cover

Bottom Clear Cover CC_b = 0.050 m
 Side Clear Cover CC_s = 0.050 m
 Pile in Pile Cap PC_p = 0.075 m

Loading applied at top of cap

Load Case	F_x (kN)	F_y (kN)	F_z (kN)	M_x (kNm)	M_y (kNm)	M_z (kNm)
1	0.012	-2.683	1.991	0.000	0.000	0.000
2	-0.004	-0.292	0.266	0.000	0.000	0.000
3	0.000	-0.000	0.000	0.000	0.000	0.000
4	0.012	-2.683	1.991	0.000	0.000	0.000
5	0.008	-2.975	2.258	0.000	0.000	0.000
6	0.010	-4.223	3.214	0.000	0.000	0.000
7	-1.086	-7.970	2.559	0.000	0.000	0.000
101	-1.049	-20.826	12.281	0.000	0.000	0.000
104	-1.049	-20.826	12.281	0.000	0.000	0.000
201	-1.468	-29.156	17.193	0.000	0.000	0.000
202	-1.258	-24.991	14.737	0.000	0.000	0.000

Two Pile Combination cannot resist transverse moment, hence for shear, punching, BM calculations transverse moment is ignored. So for pile cap parallel to X, $M_x=0, F_z=0$. And for pile cap parallel to Z, $M_z=0, F_x=0$.

PILE CAP DESIGN CALCULATION

Pile Reactions

Total pile number $N = 2$

Pile No.	Arrangement		Reaction		
	X (m)	Y (m)	Axial (kN)	Lateral (kN)	Uplift (kN)
1	-0.500	0.000	-3.354	0.996	0.000
2	0.500	0.000	-3.361	0.996	0.000

Reinforcement Calculation

Maximum bar size allowed # 16

Bending Moment At Critical Section = -6.579 kNm (Along Length)

Pile Cap Thickness $t = 0.30$ m

Selected bar size # 12

Selected bar spacing = 55 mm

Pile Cap Thickness Check

Critical Load Case : 1

Check for One-way Shear (Along Length)

Pile No.	Shear Force $x_1 - x_1$ (kN)
1	-3.354
2	-3.361

Calculation of Maximum Bar Size

Selected maximum bar size # 16

Bar diameter corresponding to max bar size (d_b) = 16.000 mm

$$\text{devlength 1} = \frac{0.87 \times d_b \times f_y}{4 \times \beta \times \sqrt{f_{cu}}} = 0.000 \text{ m}$$

$$\text{devlength 2} = 8.0 \times d_b = 0.128 \text{ m}$$

$$\text{Chosen development length } (l_{db}) = \max(\text{devlength 1, devlength 2, } 6") = 0.152 \text{ m}$$

$$\text{Available development length } (l_d) = \frac{(\text{Pile cap Length} - \text{Pedestal Dia})}{2 - \text{Cover}} = \text{m}$$

Selection of Reinforcement

Along Length

Critical Load Case : 201

$$\text{Governing moment } (M_u) = -6.579 \text{ kNm}$$

$$\begin{aligned} d\beta &= 0.85, \text{ if } F_c \leq 4000 \\ &= 0.65, \text{ if } 4000 < F_c \leq 8000 \\ &= 0.85 - \frac{0.05 \times (F_c - 4000)}{1000} = \text{ , if } F_c > 8000 \end{aligned}$$

$$\text{Maximum Reinforcement Ratio } (R_{\max}) = \frac{0.85 \times 87000 \times \alpha \beta \times F_c}{1.33 \times F_y \times (87000 + F_y)} = 0.019$$

$$\text{Calculated Reinforcement Ratio } (R) = 0.003$$

$$0.0018 \leq R \leq R_{\max}, \text{ R is accepted}$$

$$\text{Minimum spacing allowed } (S_{\min}) = 1.5 + d_b = 50 \text{ mm}$$

$$\text{Selected spacing } (S) = 55 \text{ mm}$$

$S_{\min} \leq S \leq 18$ inch and selected bar size < The reinforcement is accepted.
selected maximum bar size...

Along Width

$$\text{Governing moment } (M_u) = 0.000 \text{ kNm}$$

$$\alpha \beta = 0.85, \text{ if } F_c \leq 4000 \text{ psi}$$

$$= 0.65, \text{ if } 4000 \text{ psi} < F_c \leq 8000 \text{ psi}$$

$$= 0.85 - \frac{0.05 \times (F_c - 4000)}{1000} = \quad , \text{ if } F_c > 8000 \text{ psi}$$

$$\text{Maximum Reinforcement Ratio } (R_{\max}) = \frac{0.85 \times 87000 \times \alpha \beta \times F_c}{1.33 \times F_y \times (87000 + F_y)} = 0.000$$

$$\text{Calculated Reinforcement Ratio } (R) = 0.000$$

$$0.0018 \leq R \leq R_{\max}, \text{ R is accepted}$$

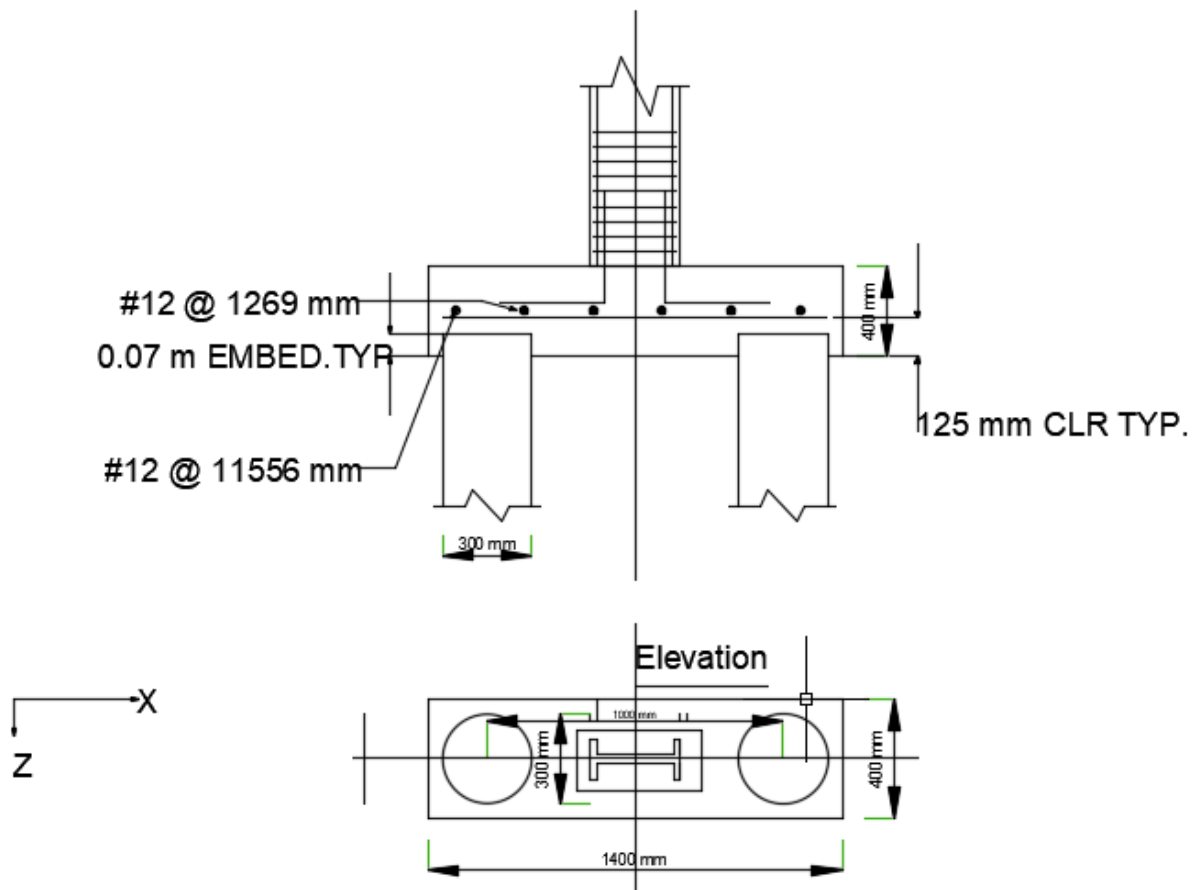
$$\text{Minimum spacing allowed } (S_{\min}) = 1.5 + d_b = 0 \text{ mm}$$

$$\text{Selected spacing } (S) = 0 \text{ mm}$$

$S_{\min} \leq S \leq 18$ inch and selected bar size < The reinforcement is
selected maximum bar size... accepted.

Selection of Top Reinforcement

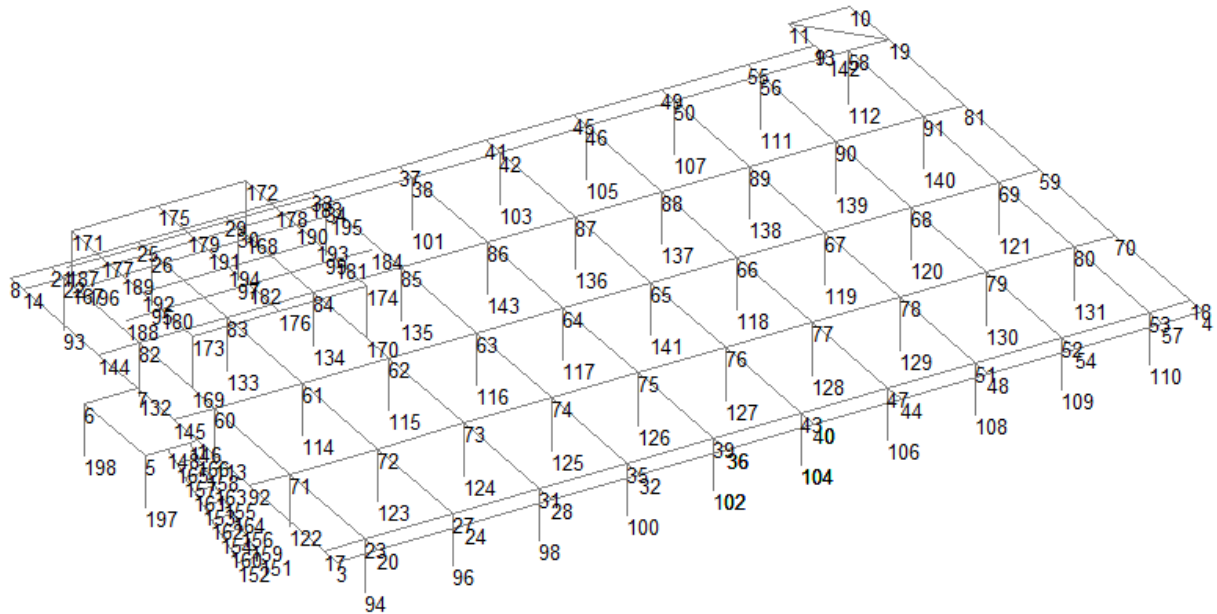
Top reinforcement is provided same as bottom reinforcement



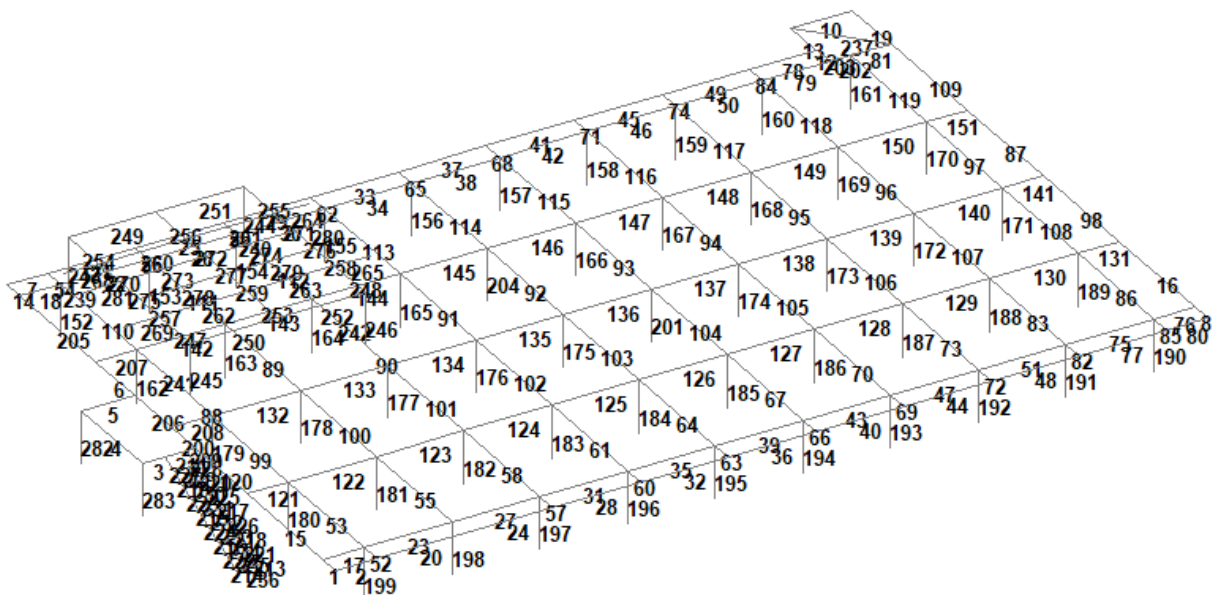
7. Annexes

7.1. Plots of Model (ELEPA)

Joint Names

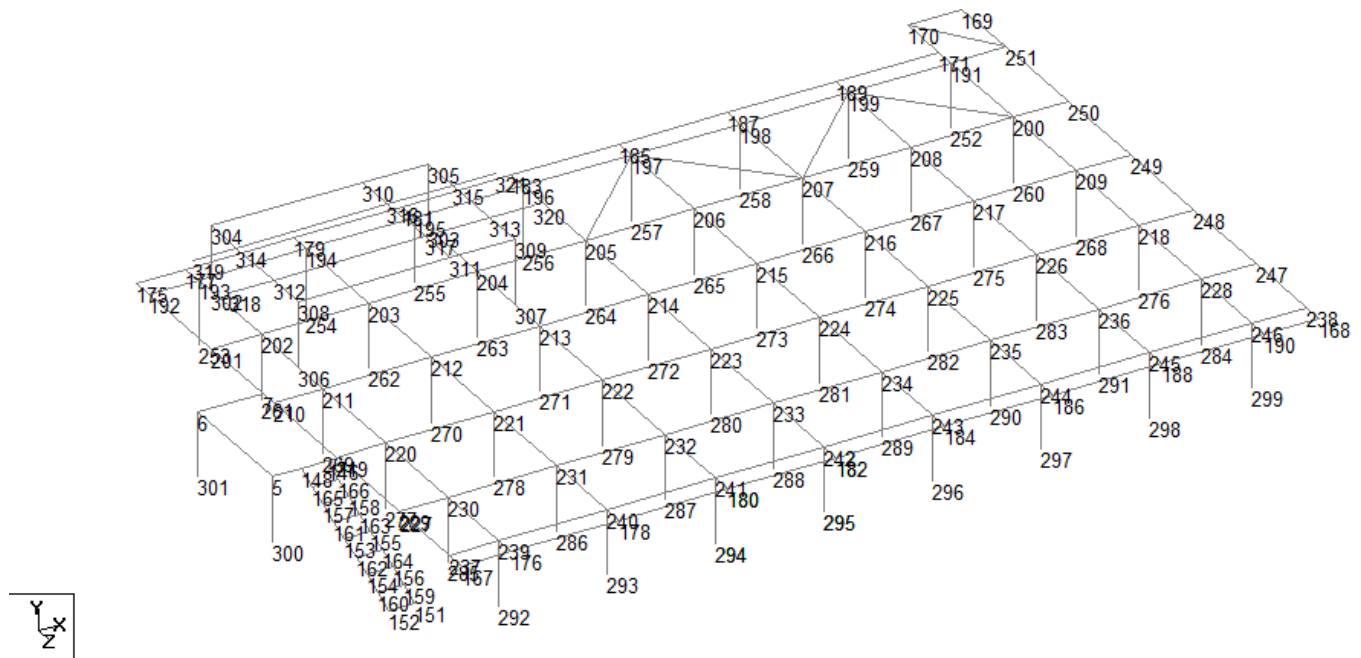


Member Names

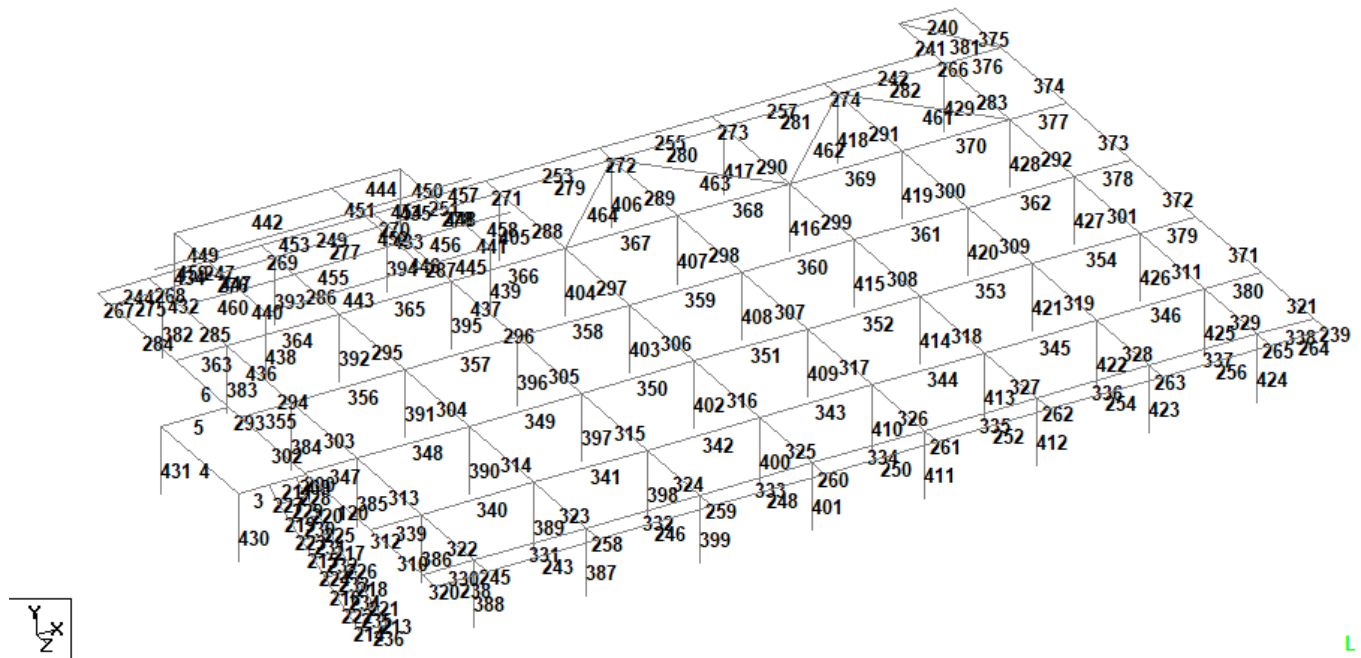


7.2. Plots of Model (BUBOUWE)

Joint Names



Member Names



L

7.3. Input file (BUBOUWE)



BB Input.pdf

7.4. Output files



Output Report.pdf

7.5. Input file (ELEPA)



Elepa Input.pdf

7.6. Output files



Elepa Output.pdf

